Last Time:

- V v.s., then $\dim V$ (dimension) $:= \begin{cases} \infty & \text{if } \nexists \text{ fin basis} \\ \#\beta & \text{if } \exists \text{ fin basis } \beta \end{cases}$

  (V is called finite dimensional (f.d.) in this case)

- Basis Extension Thm (BET): V f.d. v.s., $\gamma \subseteq V$ l.i., then $\exists \beta \supseteq \gamma$ in V that's a basis.

- $W \subseteq V$ subsp, V f.d. Then $\dim W \leq \dim V$, and $\dim W = \dim V \iff W = V$.

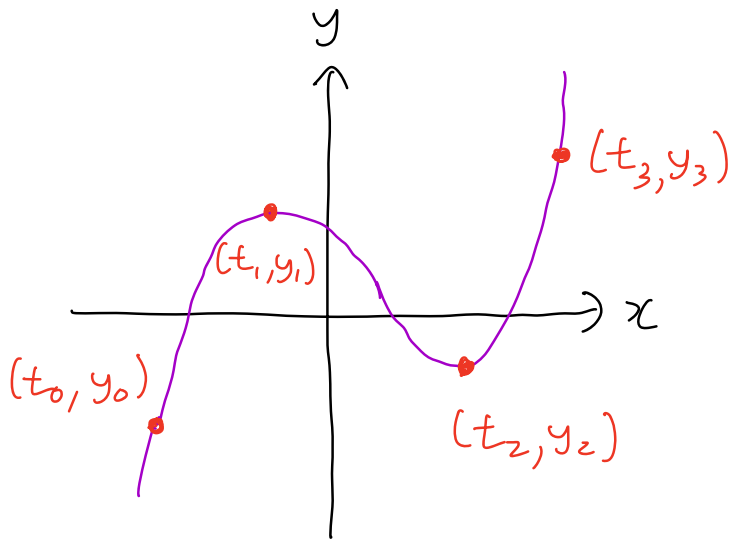
Today:

(1) Continue §1.4 Basis and Dimension (Application)

(2) Start §2.1 Linear Maps: ker, im, isom

Application: Lagrange Interpolation Formula

Motivation: Given data pts $(t_0, y_0), \dots (t_n, y_n) \in \mathbb{R}^2$ s.t. $t_0, \dots, t_n$ distinct, how do we find a poly whose graph passes thru those data pts?



Strategy: Find a suitable basis of $\mathbb{R}[x]_{\leq n}$ and

constr a specific l.c. as the desired poly.

Def 1.62: Given $t_0, \dots, t_n \in \mathbb{R}$ distinct, assoc associated Lagrange polys are of the form

$$f_i(x) := \frac{(x-t_0)\cdots(x-t_{i-1})(x-t_{i+1})\cdots(x-t_n)}{(t_i-t_0)\cdots(t_i-t_{i-1})(t_i-t_{i+1})\cdots(t_i-t_n)}$$

$$= \prod_{\substack{0\le j\le n \\ j\neq i}} \frac{(x-t_j)}{(t_i-t_j)} \quad \underbrace{\text{deg } n \text{ poly}}_{\in \mathbb{R}[x]_{\le n}} \quad \forall\, 0 \le i \le n.$$

(Observe that $f_i(t_j) \overset{(\star)}{=} \delta_{ij}$)

Prop 1.63: $f_0, \dots, f_n \in \mathbb{R}[x]_{\le n}$ Lagrange polys assoc w/ $t_0, \dots, t_n \in \mathbb{R}$. Then $f_0, \dots, f_n$ is a basis.

Pf: $\dim \mathbb{R}[x]_{\le n} = n+1$ by Ex 1.58(3), so

Want to Show (WTS): $f_0, \dots, f_n$ l.i., b/c l.i. $\Rightarrow f_0, \dots, f_n$ basis by Cor 1.59(2).

To check l.i., consider equation (eqn) $0 = \sum_{i=0}^{n} a_i f_i$

$\Rightarrow \forall j = 0, \dots, n$, ($x = t_j$) $0 = \sum_{i=0}^{n} a_i f_i(t_j) \overset{(\star)}{=} \sum_{i=0}^{n} a_i \delta_{ij} = a_j$

$\Rightarrow f_0, \dots f_n$ l.i. by def. $\square$

Cor 1.64 (Lagrange Interpolation Formula (LIF)):
Given $t_0, \dots, t_n \in \mathbb{R}$ distinct and $y_0, \dots, y_n \in \mathbb{R}$,
$\exists!\ p \in \mathbb{R}[x]_{\le n}$ s.t. $p(t_i) = y_i \ \ \forall i$.

Pf: **Existence**: Let $p = \sum_{i=0}^{n} y_i f_i$, where $f_0, \dots, f_n$ Lagrange polys assoc w/ $\{t_0, \dots t_n\}$. Then

$$\forall\, i = 0, \dots, n,\quad p(t_i) = \sum_{j=0}^{n} y_j f_j(t_i) \overset{(\star)}{=} \sum_{j=0}^{n} y_j \delta_{ji} = y_i. \ \blacksquare$$

Uniqueness: B/c $f_0, \dots, f_n$ is a basis by Prop 1.62, every $g \in \mathbb{R}[x]_{\le n}$ has a ! l.c. $c_0 f_0 + \cdots + c_n f_n$ by Prop 1.48. Similar to prev argument, $(c_0 f_0 + \cdots + c_n f_n)(t_i) = c_i \ \ \forall i = 0, \dots, n$, so conds $g(t_i) = y_i \ \forall i \Rightarrow c_i = y_i \ \forall i \Rightarrow g = p$. $\square$

Ex 1.65: Find a poly whose graph passes thru $(1,8), (2,5), (3,-4) \in \mathbb{R}^2$.

Sol: Let $t_0 = 1$, $t_1 = 2$, $t_2 = 3$. Then assoc Lagrange polys f_0, f_1, f_2 are

$$f_0(x) = \frac{(x-t_1)(x-t_2)}{(t_0-t_1)(t_0-t_2)} = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{1}{2}(x^2 - 5x + 6)$$

$$f_1(x) = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -(x^2-4x+3)$$

$$f_2(x) = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{1}{2}(x^2-3x+2)$$

Then desired poly is

$$p(x) = 8f_0(x) + 5f_1(x) - 4f_2(x)$$

$$= 4(x^2-5x+6) - 5(x^2-4x+3) - 2(x^2-3x+2)$$

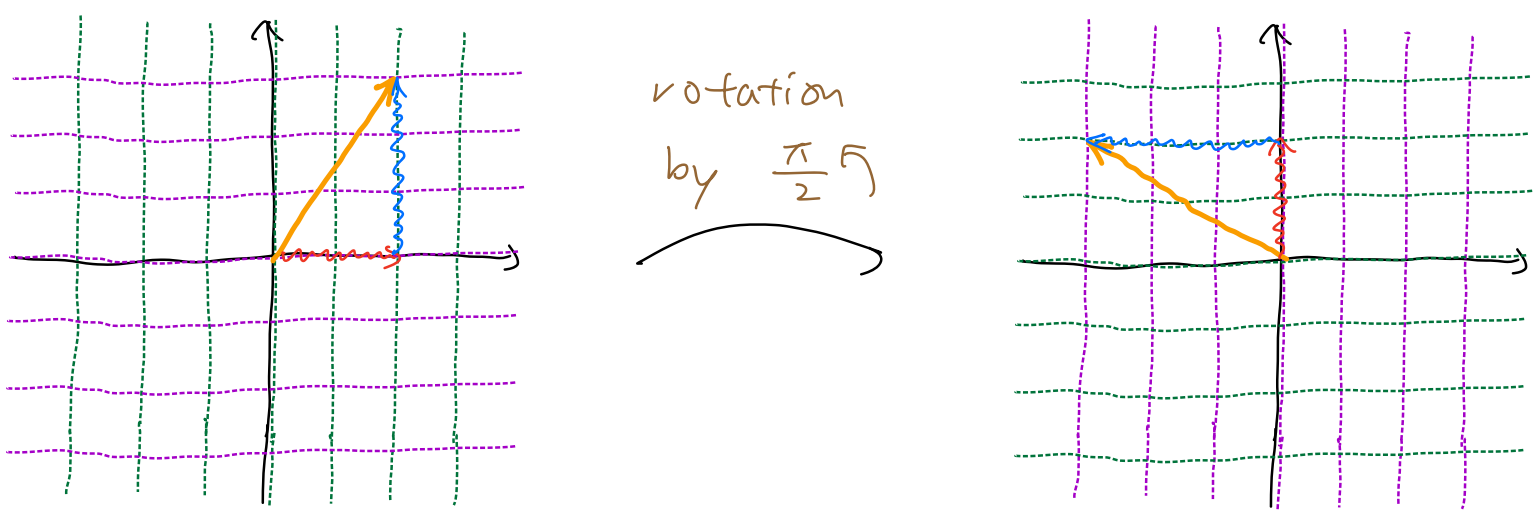
$$= \boxed{-3x^2+6x+5}$$

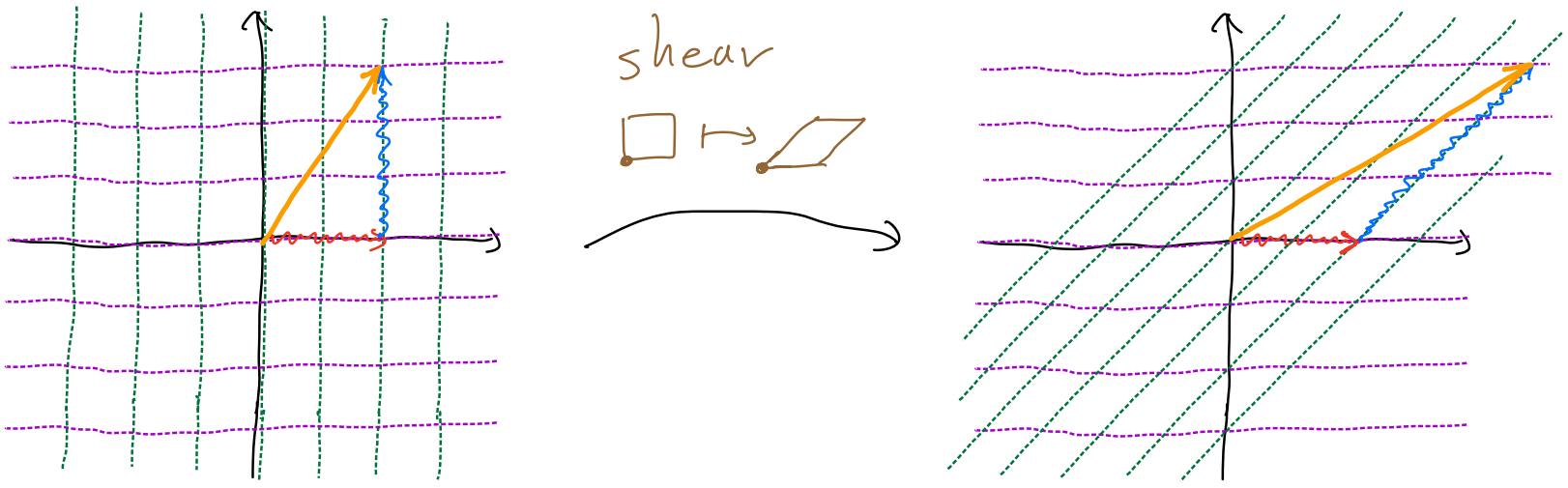
Unit 2: Linear Transformations

This unit is about "special" fcns btwn v.s.

Motivation: So far, we studied properties of v.s. What about fcns/maps btwn v.sps?

Look at some photoshop operations:





They fix origin, send any line to a line, and send any parallelogram to a parallelogram. Using  , we want fcns that preserve addition & scalar-mult.

Set Theory Warm-Up: given a fcn/map $f: X \to Y$, (X: domain, Y: codomain)

$\operatorname{im}(f) = \underset{(FIS)}{\text{range}}(f) = f(X) = \{f(x) \in Y : x \in X\} \subseteq Y$ is called range/image of f. Given $Z \subseteq Y$,

$f^{-1}(Z) := \{x \in X : f(x) \in Z\}$ is the preimage of $Z \subseteq Y$ in X.

- when f is 1-1 & onto, $f^{-1}: Y \to X$ exists, and
- $f^{-1}(Z)$ is the img of $Z \subseteq Y$ via f^{-1}.

§2.1 Linear Maps: ker, im, and isom

As in Motivation, let's consider fcns btwn vector spaces that preserve $+$ & $\cdot$:

Def 2.1: V, W v.s. A fcn map $T: V \to W$ is a lin transf linear transformation (or lin map / hom homomorphism) if $\forall v_1, v_2 \in V$,

(1) $T(v_1 +_V v_2) = T(v_1) +_W T(v_2)$ $\left(T(v_1+v_2) = T(v_1)+T(v_2)\right)$

(2) $T(c \cdot_V v_1) = c \cdot_W T(v_1) \quad \forall c \in \mathbb{R}$. $\left(T(cv_1) = cT(v_1)\right)$

The set of lin maps $V \to W$ is called $\mathcal{L}(V,W)$.

(When $V = W$, we say $\mathcal{L}(V) := \mathcal{L}(V,V)$)

Rmk 2.2: $T \in \mathcal{L}(V,W)$. Then

(1) $T(0_V) = 0_W$ $\quad (T(0) = 0)$

(2) $T(-v) = -T(v) \quad \forall v \in V$

Pf: (1): Fix any $v \in V$. Then

$$T(0_V) = T(0 \cdot_V v) = 0 \cdot_W T(v) = 0_W.$$

(Prop 1.17(1), Def 2.1, Prop 1.17(1)) □

(2): $\forall v \in V$,

$$T(-v) = T((-1) \cdot_V v) = (-1) \cdot_W T(v) = -T(v).$$

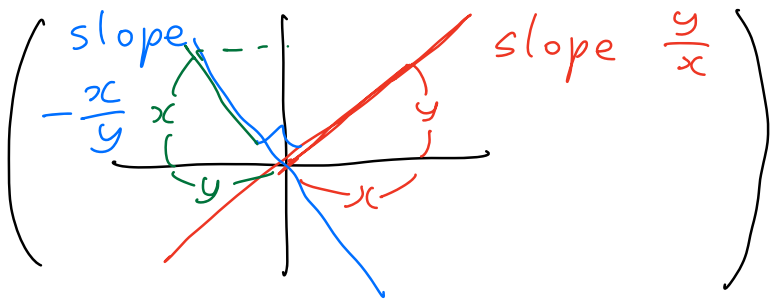
(Def 2.1, Prop 1.17(2), Prop 1.17(2)) □

Ex 2.3:

(1) Rotation by $\frac{\pi}{2}$ CCW in $\mathbb{R}^2$ is def by

$$R : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \longmapsto \begin{bmatrix} -y \\ x \end{bmatrix}$$



This is a lin map.

Pf: Next Class!